Annotation and Annotation Resources

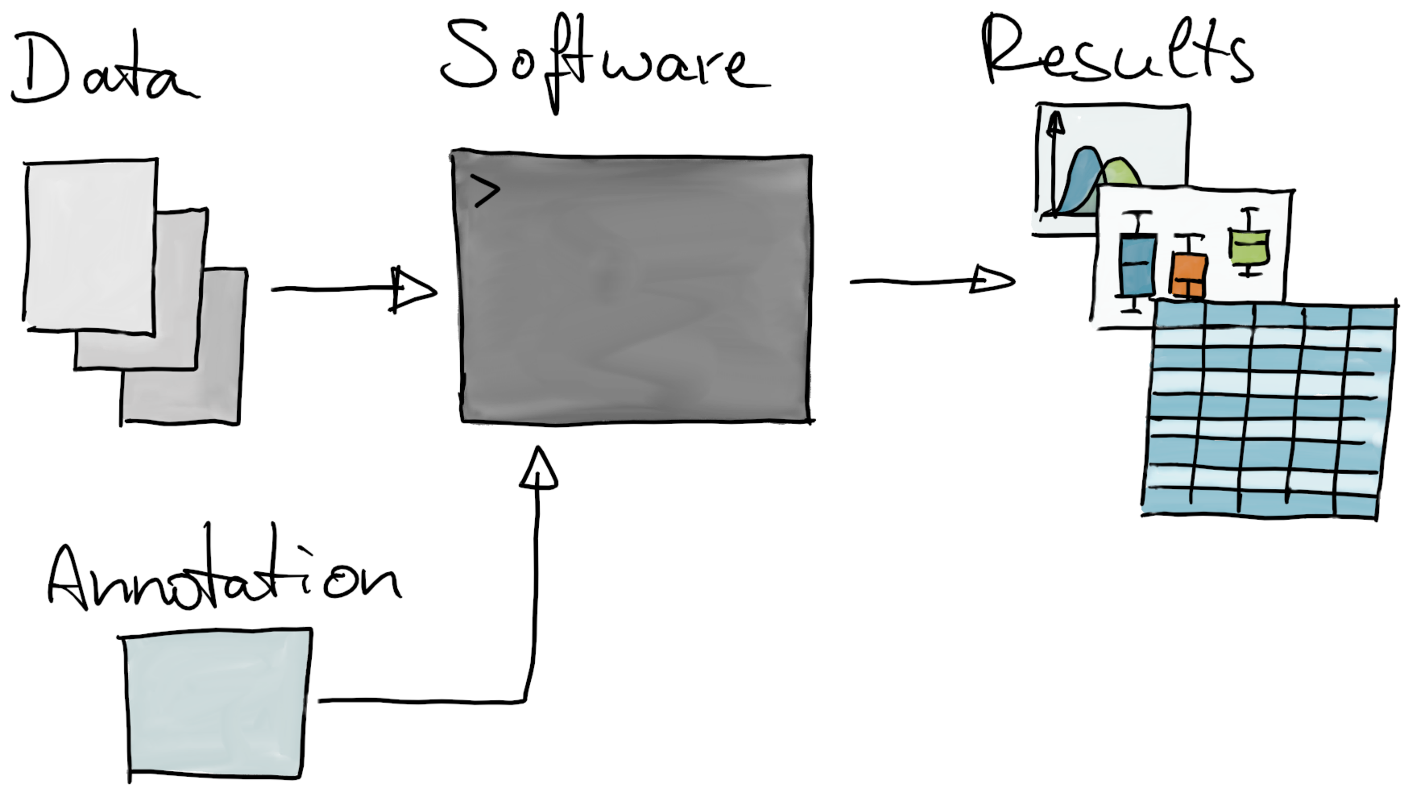
^ with content from Lori Shepherd Kern, Martin Morgan and James W. MacDonald

CSAMA 2026

Johannes Rainer
Institute for Biomedicine, Eurac Research

Introduction

Data analysis



- *Annotation*: external data fed into an analysis to make *sense* of quantified entities.

Annotation

- put a (new or different) label on an entity
- *entity*: something measured with an assay (RNA, molecule, cell)
- *label*: commonly used (standard) identifier (e.g. gene name)
- sounds trivial, but is very important task in computational biology
- examples:
  - map positions on the genome to transcripts
  - map genes to biological pathways
  - annotate cell types based on presence of marker genes
- 🎯 enable or assist in interpretation; make biological sense of assay data.

In this presentation

- 1️⃣ provide some general information on annotation
- 2️⃣ give you an overview on available annotation resources
- 3️⃣ short use cases (with code)

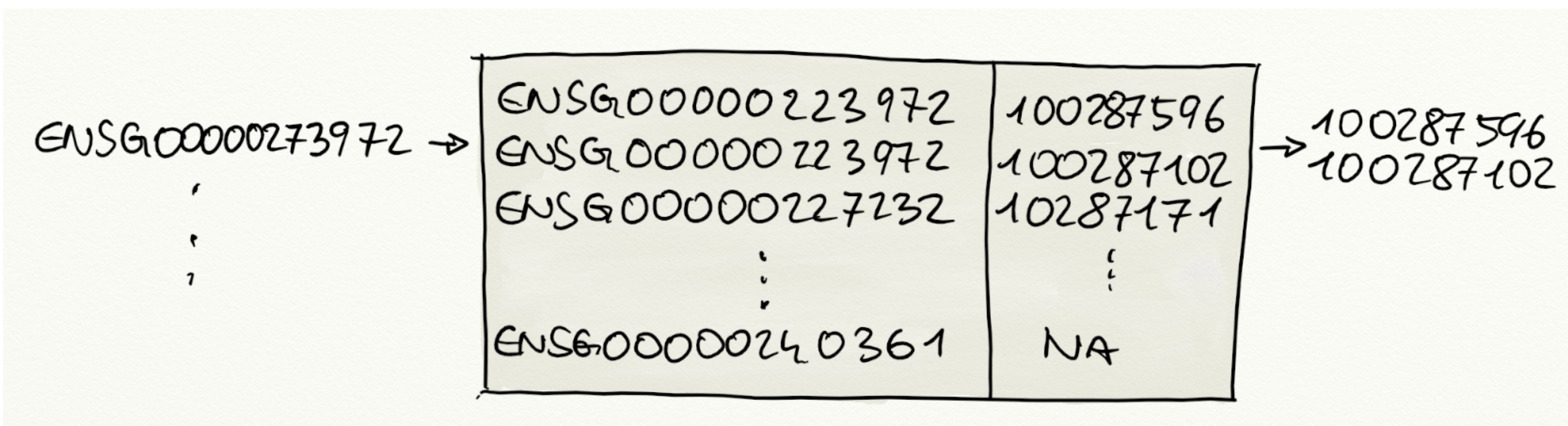
1️⃣ General information on annotation

What do we need for annotation?

- 1️⃣ an **annotation resource**: reference information on entities or a mapping of one type of identifiers to another.
- 2️⃣ a **rule** that defines how to assign labels to entities
  - annotation through similarity: sequence similarity, spectral similarity
  - direct mapping

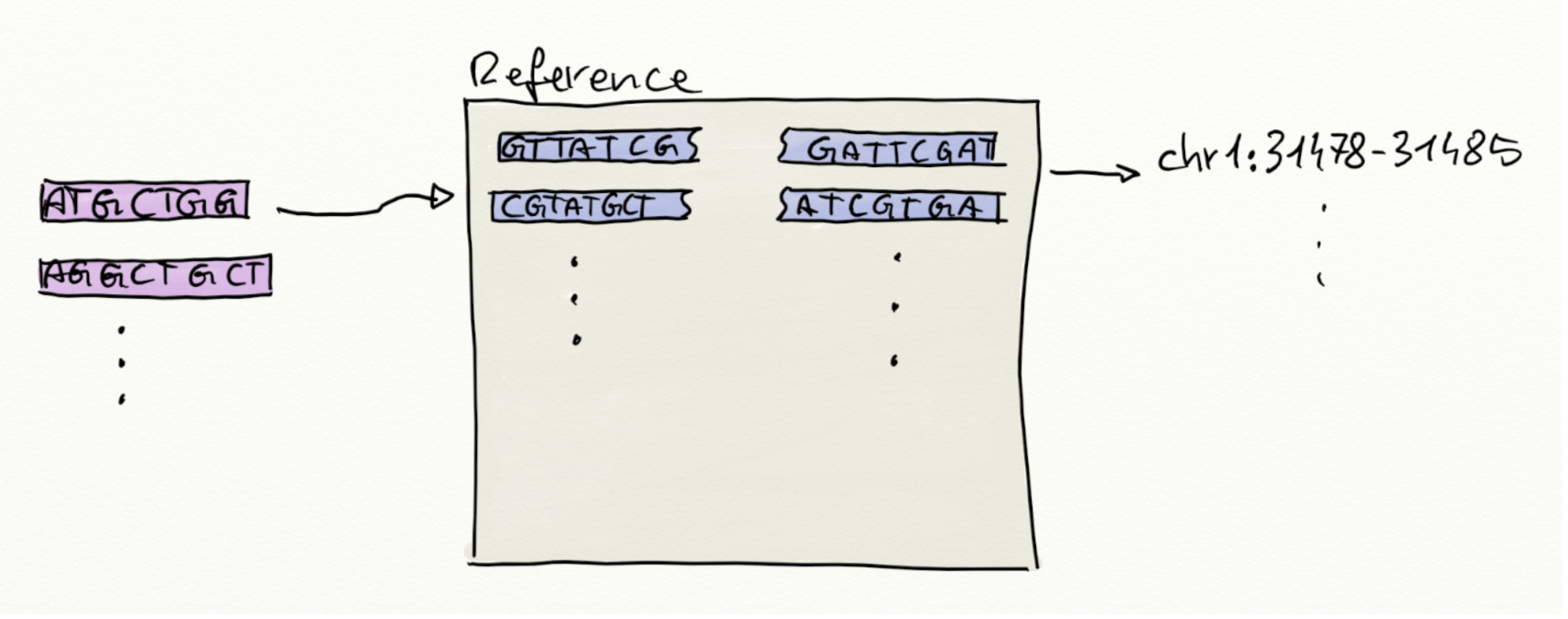
What do we need for annotation?

- 1️⃣ an **annotation resource**: reference information on entities or a mapping of one type of identifiers to another.
- 2️⃣ a **rule** that defines how to assign labels to entities
- *examples*
  - map between gene identifiers (Ensembl, NCBI, ...)



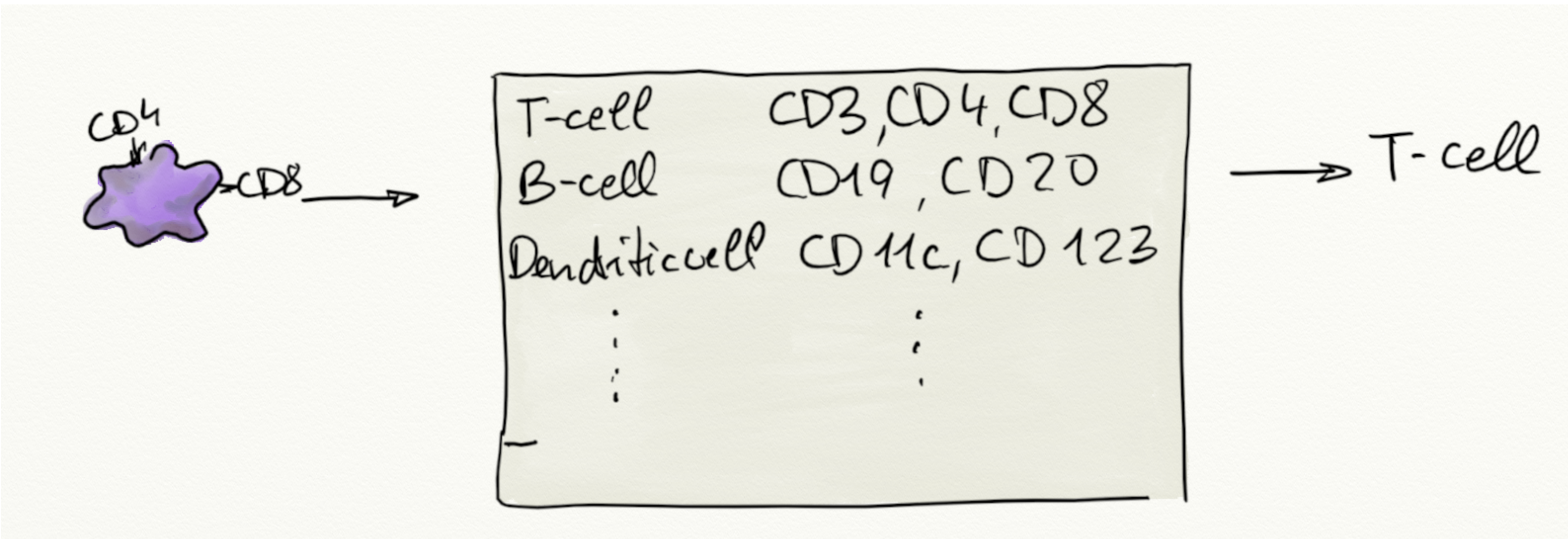
What do we need for annotation?

- 1️⃣ an **annotation resource**: reference information on entities or a mapping of one type of identifiers to another.
- 2️⃣ a **rule** that defines how to assign labels to entities
- *examples*
  - annotate through sequence similarity: map short reads to genome



What do we need for annotation?

- 1️⃣ an **annotation resource**: reference information on entities or a mapping of one type of identifiers to another.
- 2️⃣ a **rule** that defines how to assign labels to entities
- *examples*
  - cell types through presence of marker genes



Reproducibility

In order to guarantee **reproducibility** of annotation:

✅ annotation resource has to be **versioned** and **findable**

🙅 dynamic annotation resources (daily updates) without the possibility to access a specific version

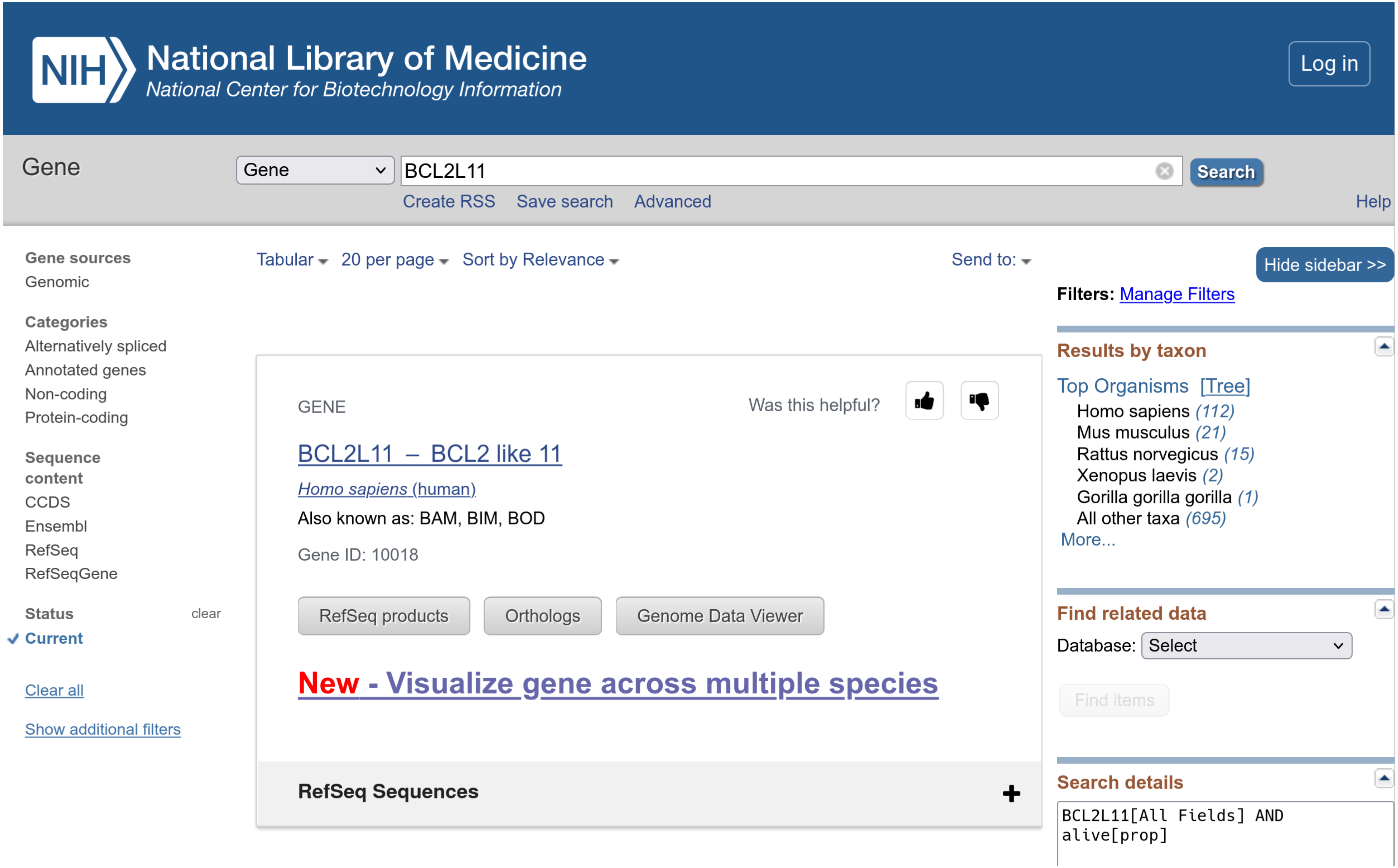
✅ the **same version** should be used throughout the **whole** analysis

🙅 using a different genome release for RNA-seq alignment and gene counting.

Standardization and metadata

- **Common nomenclature is required**
- It’s great if you can assign a name to an entity - but also other researchers should understand what you mean.

- *example*: synonyms



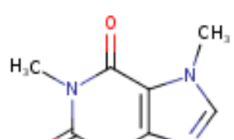
- Synonyms: aliases or previous names of entities.

Standardization and metadata

- 🙀 … it can always be worse …

Showing metabocard for Caffeine (HMDB0001847)

Record Information	
Version	5.0
Status	Detected and Quantified
Creation Date	2006-02-16 08:53:46 UTC
Update Date	2023-02-21 17:15:49 UTC
HMDB ID	HMDB0001847
Secondary Accession Numbers	• HMDB01847

Metabolite Identification	
Common Name	**Caffeine**
Description	Caffeine is a methyl xanthine alkaloid that is also classified as a purine. Formally, caffeine belongs to the class of organic compounds known as xanthines. These are purine derivatives with a ketone group conjugated at carbons 2 and 6 of the purine moiety. Caffeine is chemically related to the adenine and guanine bases of deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). It is found in the seeds, nuts, or leaves of a number of plants native to Africa, East Asia and South America and helps to protect them against predator insects and to prevent germination of nearby seeds. The most well-known source of caffeine is the coffee bean. Caffeine is the most widely consumed psychostimulant drug in the world. 85% of American adults consumed some form of caffeine daily, consuming 164 mg on average. Caffeine is mostly is consumed in the form of coffee. Caffeine is a central nervous system stimulant that reduces Read more...
Structure	

Why? annotation resource was compiled from community provided information.

- **Standardization and common nomenclature is important!**
- Standardization is important for human consumers, but even more so for computational use (including AI).

2 Annotation resources

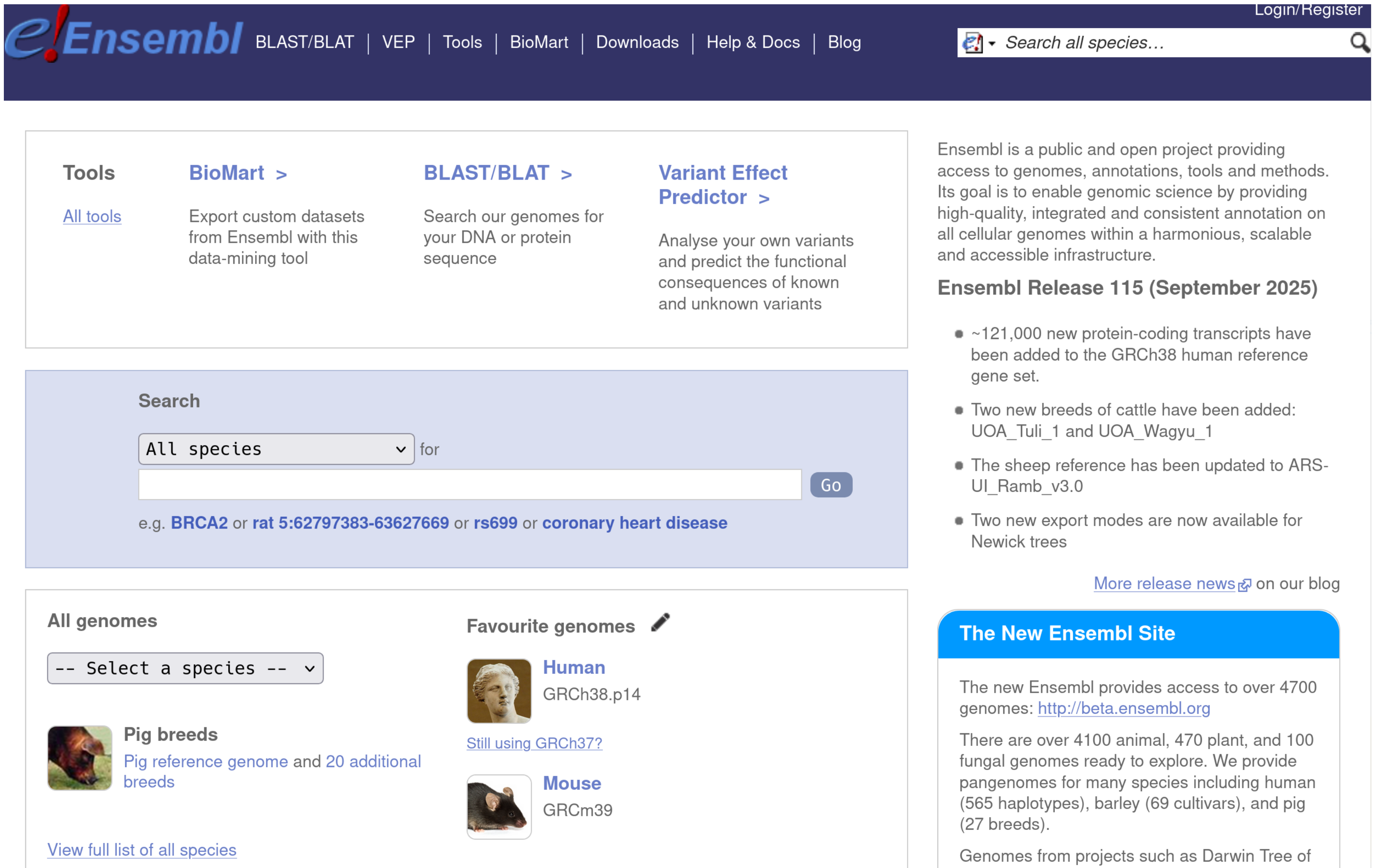
Annotation resources

- Where can you get annotations from?

Annotation resources

For genes

- NCBI, Ensembl, …



Annotation resources

For proteins

- UniProt, Ensembl, ...

UniProt Tools ▾ UniProtKB ▾ BCL2L11 Advanced | List Search Help

Please read our help page, view affected entries and proteomes, or contact us with any questions.

Status

Reviewed (Swiss-Prot) (129)

Unreviewed (TrEMBL) (2,845)

Popular organisms

Human (68)

Rat (32)

Mouse (31)

Bovine (24)

Zebrafish (7)

Taxonomy

Filter by taxonomy

Group by

Taxonomy

Keywords

Gene Ontology

Enzyme Class

UniProtKB 2,974 results

or search "BCL2L11" as a Gene Name or Protein Name

Tools ▾ Download (3k) Add View: Cards Table Customize columns Share ▾

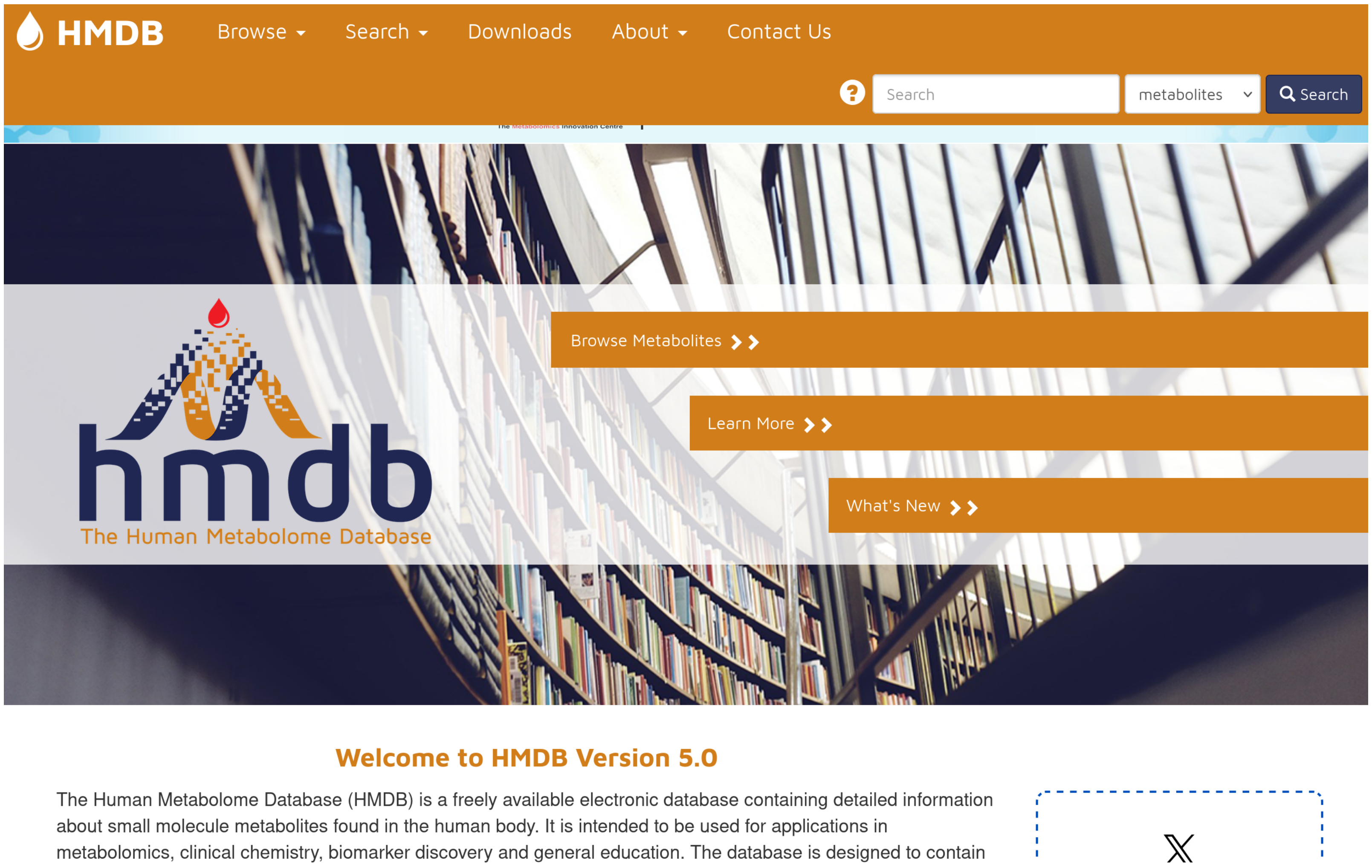
Entry		Entry Name	Protein Names	Gene Names	Organism	Len
O43521		B2L11_HUMAN	**Bcl-2-like protein 11**[...]	**BCL2L11**, BIM	Homo sapiens (Human)	198
O54918		B2L11_MOUSE	**Bcl-2-like protein 11**[...]	**Bcl2l11**, Bim	Mus musculus (Mouse)	196
P63167		DYL1_HUMAN	**Dynein light chain 1, cytoplasmic**[...]	**DYNLL1**, DLC1, DNCL1, DNCLC1, HDLC1	Homo sapiens (Human)	89 A
Q07820		MCL1_HUMAN	**Induced myeloid leukemia cell differentiation protein Mcl-1**[...]	**MCL1**, BCL2L3	Homo sapiens (Human)	350
Q07812		BAX_HUMAN	**Apoptosis regulator BAX**[...]	**BAX**, BCL2L4	Homo sapiens (Human)	192
O88498		B2L11_RAT	**Bcl-2-like protein 11**[...]	**Bcl2l11**, Bim, Bod	Rattus norvegicus (Rat)	

Help

Annotation resources

For small molecules

- HMDB, MassBank, …



MassBank
Features
Quick Search
Latest News
Funding

MassBank

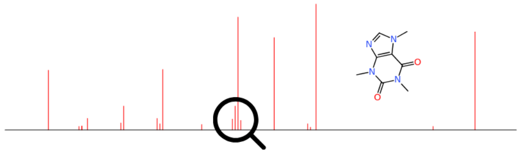
High Quality Mass Spectral Database

Welcome to MassBank, an open-source mass spectral library for the identification of small chemical molecules of metabolomics, exposomics and environmental relevance.

Features

The MassBank system provides a variety of functionalities to access, filter and search for data within its knowledge base. The following key features are available.

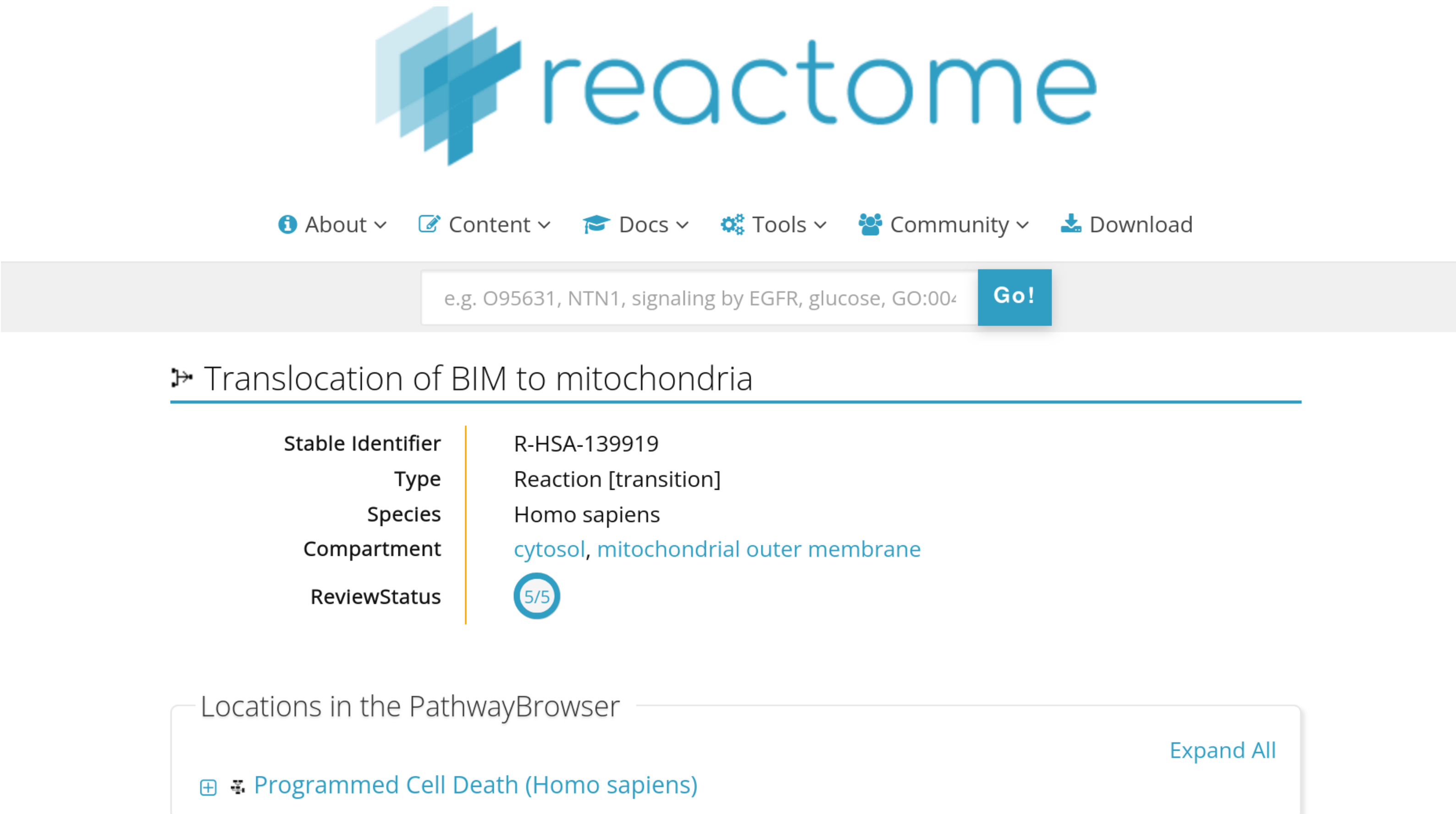
- **Compound Search**
- **Spectral Search**
- **Structure Search**
- **MassBank API**



Annotation resources

Pathways

- reactome, wikipathways, KEGG



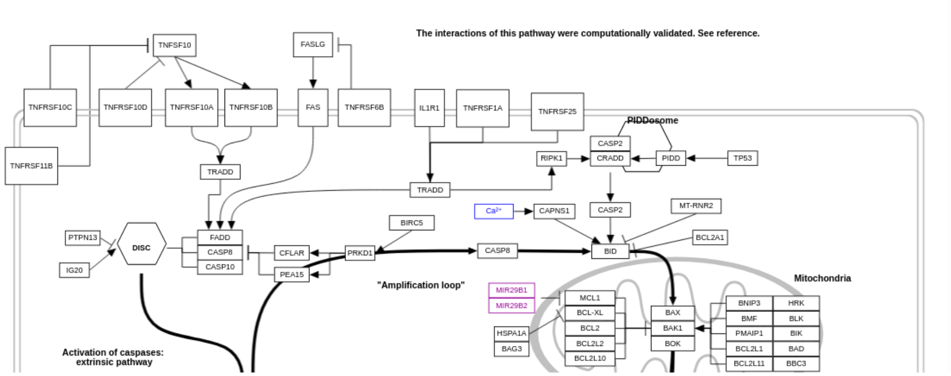
Annotation resources

Pathways

- reactome, wikipathways, KEGG

Apoptosis modulation and signaling (WP1772)

Homo sapiens



Annotation resources

Pathways

- reactome, wikipathways, KEGG

Homo sapiens (human): 10018

Help

Entry	10018 CDS T01001
Symbol	BCL2L11, BAM, BIM, BOD
Name **KO**	(RefSeq) bcl-2-like protein 11 isoform 1 K16341 Bcl-2-like protein 11
Organism	hsa Homo sapiens (human)
Pathway	hsa01521 EGFR tyrosine kinase inhibitor resistance hsa04068 FoxO signaling pathway hsa04151 PI3K-Akt signaling pathway hsa04210 Apoptosis hsa04215 Apoptosis - multiple species hsa04932 Non-alcoholic fatty liver disease hsa05169 Epstein-Barr virus infection hsa05200 Pathways in cancer hsa05206 MicroRNAs in cancer hsa05210 Colorectal cancer
Network	nt06161 Human immunodeficiency virus 1 (HIV-1) nt06165 Epstein-Barr virus (EBV) nt06166 Human papillomavirus (HPV) nt06168 Herpes simplex virus 1 (HSV-1) nt06170 Influenza A virus (IAV) nt06231 Apoptosis (cancer) nt06260 Colorectal cancer nt06267 Small cell lung cancer nt06463 Parkinson disease

- 😒 some annotation resources use their own data/file format making it cumbersome to include them in data analysis workflows.

Bioconductor annotation resources

- Format standardized to simplify integration into the analysis (big thanks to the maintainers! 🙌)
- *AnnotationDbi* package defines a common interface to extract information from annotation resources: `mapIds()` and `select()` functions.

Bioconductor annotation resources

- 💍 one to get them all: *AnnotationHub*

- Central registry for annotation resources.

```r
library(AnnotationHub)

#' Synchronize and cache AnnotationHub information locally
ah ← AnnotationHub()
#' List all available resources
ah
```

```
AnnotationHub with 69129 records
# snapshotDate(): 2026-04-23
# $dataprovider: Ensembl, BroadInstitute, UCSC, Haemcode, FANTOM5,DLRP,IUPHA ...
# $species: Homo sapiens, Mus musculus, Drosophila melanogaster, Rattus norv ...
# $rdataclass: GRanges, TwoBitFile, BigWigFile, EnsDb, Rle, ChainFile, SQLit ...
# additional mcols(): taxonomyid, genome, description,
#   coordinate_1_based, maintainer, rdatadateadded, preparerclass, tags,
#   rdatapath, sourceurl, sourcetype
# retrieve records with, e.g., 'object[["AH5012"]]'

            title
  AH5012   | Chromosome Band
  AH5013   | STS Markers
  ...        ...
  AH122185 | Data.table for MeSH (Qualifier)
  AH122186 | Data.table for MeSH (SCR)
```

- Resources come in variety of different formats and data types.

```r
#' List data class
table(ah$rdataclass)
```

```
                 AAStringSet                          BigWigFile
                           1                               10247
                     biopax                            ChainFile
                           9                                1115
                   character                              CompDb
                          10                                   8
                  data.frame data.frame, DNAStringSet, GRanges
                          57                                   3
                  data.table                               EnsDb
                          25                                5272
                      FaFile                             GRanges
                           3                               30545
                      igraph                       Inparanoid8Db
                           2                                 268
                      JASPAR                                list
                           3                                  71
                      MSnSet                            mzRident
                           1                                   1
                      OrgDb                                  Rda
```

- Resource can be a reference to an external file

```r
query(ah, c("ensembl", "TwoBitFile"))
```

```
AnnotationHub with 17681 records
# snapshotDate(): 2026-04-23
# $dataprovider: Ensembl
# $species: Mus musculus, Homo sapiens, Danio rerio, Xiphophorus maculatus, ...
# $rdataclass: TwoBitFile
# additional mcols(): taxonomyid, genome, description,
```

```
#   coordinate_1_based, maintainer, rdatadateadded, preparerclass, tags,
#   rdatapath, sourceurl, sourcetype
# retrieve records with, e.g., 'object[["AH49592"]]'

             title
  AH49592  | Ailuropoda_melanoleuca.ailMel1.cdna.all.2bit
  AH49593  | Ailuropoda_melanoleuca.ailMel1.dna_rm.toplevel.2bit
  ...        ...
  AH107046 | Zosterops_lateralis_melanops.ASM128173v1.dna_sm.toplevel.2bit
  AH107047 | Zosterops_lateralis_melanops.ASM128173v1.ncrna.2bit
```

```
ah["AH49592"]
```

```
AnnotationHub with 1 record
# snapshotDate(): 2026-04-23
# names(): AH49592
# $dataprovider: Ensembl
# $species: Ailuropoda melanoleuca
# $rdataclass: TwoBitFile
# $rdatadateadded: 2015-12-28
# $title: Ailuropoda_melanoleuca.ailMel1.cdna.all.2bit
# $description: TwoBit cDNA sequence for Ailuropoda melanoleuca
# $taxonomyid: 9646
# $genome: ailMel1
# $sourcetype: FASTA
# $sourceurl: ftp://ftp.ensembl.org/pub/release-82/fasta/ailuropoda_melanole...
# $sourcesize: 10988959
# $tags: c("TwoBit", "ensembl", "sequence", "2bit", "FASTA")
# retrieve record with 'object[["AH49592"]]'
```

- Or a dedicated R object.
- We retrieve one such object from *AnnotationHub*.

```
query(ah, c("Hsapiens", "EnsDb"))
```

```
AnnotationHub with 28 records
# snapshotDate(): 2026-04-23
```

```
# $dataprovider: Ensembl
# $species: Homo sapiens
# $rdataclass: EnsDb
# additional mcols(): taxonomyid, genome, description,
#   coordinate_1_based, maintainer, rdatadateadded, preparerclass, tags,
#   rdatapath, sourceurl, sourcetype
# retrieve records with, e.g., 'object[["AH53211"]]'

            title
  AH53211  | Ensembl 87 EnsDb for Homo Sapiens
  AH53715  | Ensembl 88 EnsDb for Homo Sapiens
  ...        ...
  AH116860 | Ensembl 112 EnsDb for Homo sapiens
  AH119325 | Ensembl 113 EnsDb for Homo sapiens
```

```r
edb <- ah[["AH119325"]]
edb
```

```
EnsDb for Ensembl:
|Backend: SQLite
|Db type: EnsDb
|Type of Gene ID: Ensembl Gene ID
|Supporting package: ensembldb
|Db created by: ensembldb package from Bioconductor
|script_version: 0.3.10
|Creation time: Sat Oct 26 21:34:14 2024
|ensembl_version: 113
|ensembl_host: 127.0.0.1
|Organism: Homo sapiens
|taxonomy_id: 9606
|genome_build: GRCh38
|DBSCHEMAVERSION: 2.2
|common_name: human
|species: homo_sapiens
| No. of genes: 87726.
| No. of transcripts: 413674.
|Protein data available.
```

- ℹ️ data is downloaded and cached. Subsequent calls will load the data from the

Bioconductor annotation resources

- … and there are also dedicated R packages with specific annotations.

- *example*: identifiers and metadata for human genes.

```
library(org.Hs.eg.db)
org.Hs.eg.db
```

```
OrgDb object:
| DBSCHEMAVERSION: 2.1
| Db type: OrgDb
| Supporting package: AnnotationDbi
| DBSCHEMA: HUMAN_DB
| ORGANISM: Homo sapiens
| SPECIES: Human
| EGSOURCEDATE: 2026-Mar18
| EGSOURCENAME: Entrez Gene
| EGSOURCEURL: ftp://ftp.ncbi.nlm.nih.gov/gene/DATA
| CENTRALID: EG
| TAXID: 9606
| GOSOURCENAME: Gene Ontology
| GOSOURCEURL: https://current.geneontology.org/ontology/go-basic.obo
| GOSOURCEDATE: 2026-01-23
| GOEGSOURCEDATE: 2026-Mar18
| GOEGSOURCENAME: Entrez Gene
| GOEGSOURCEURL: ftp://ftp.ncbi.nlm.nih.gov/gene/DATA
| KEGGSOURCENAME: KEGG GENOME
| KEGGSOURCEURL: ftp://ftp.genome.jp/pub/kegg/genomes
```

- Most annotation resources provide mapping between gene identifiers or mapping to e.g. pathways.
- Also available: *position-relative annotations*.

Position-relative annotations

- Mapping between entities (genes, exons) and positions (on the genome, within a transcript).
- `GenomicFeatures::TxDb` resources.

Position-relative annotations

- Mapping between entities (genes, exons, **protein domains**) and positions (on the genome, within a transcript, within a **protein**).
- `GenomicFeatures::TxDb` and `ensembldb::EnsDb` resources.

Position-relative annotations

- Mapping between entities (genes, exons, protein domains) and positions (on the genome, within a transcript, within a protein).
- `GenomicFeatures::TxDb` and `ensembldb::EnsDb` resources.
- Organism and release-specific data (SQLite databases).
- Positional information: genomic position of exons

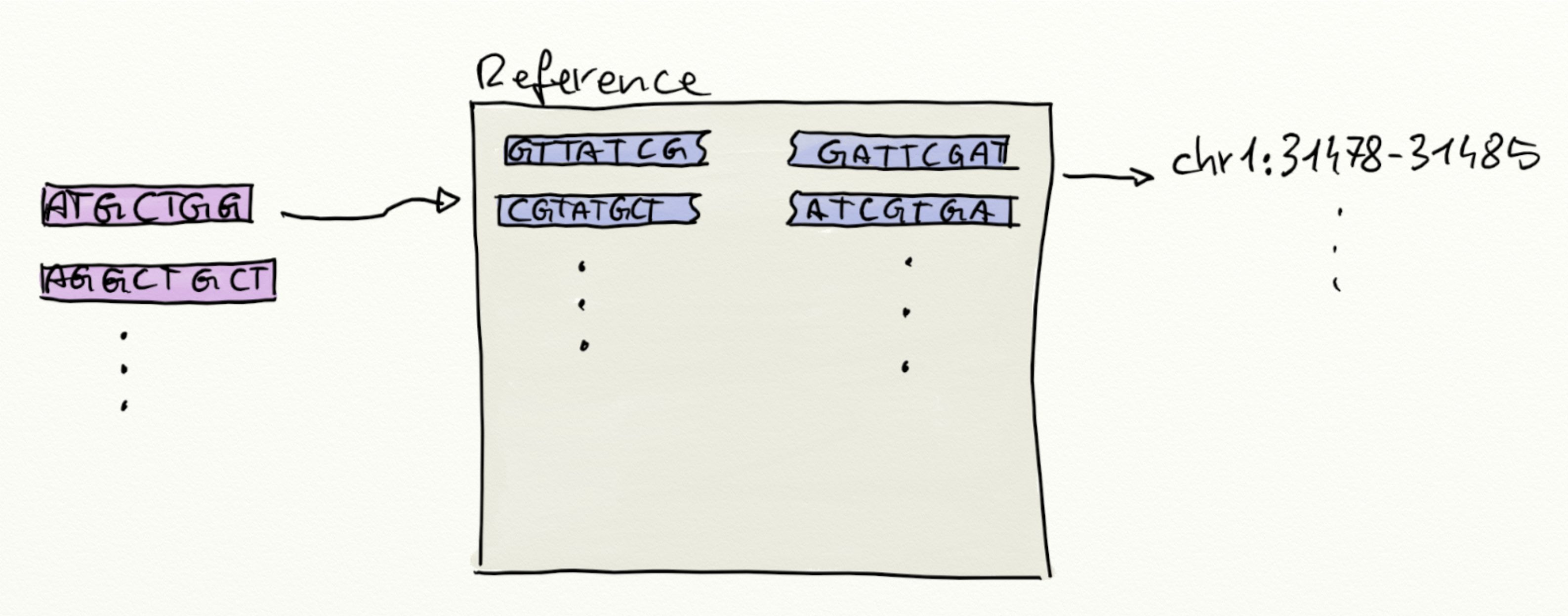
```r
#' extract exon coordinates from resource downloaded form AnnotationHub
exons(edb)
```

```
GRanges object with 1131826 ranges and 1 metadata column:
                  seqnames            ranges strand |         exon_id
                     <Rle>         <IRanges>  <Rle> |     <character>
  ENSE00004248723        1       11121-11211      + | ENSE00004248723
  ENSE00004248721        1       11125-11211      + | ENSE00004248721
              ...      ...               ...    ... .             ...
  ENSE00004286596        Y 57215558-57215594      - | ENSE00004286596
  ENSE00004286597        Y 57215558-57215634      - | ENSE00004286597
  -------
  seqinfo: 540 sequences (1 circular) from GRCh38 genome
```

ℹ️ such resources can also used to map e.g. between positions within a protein and the location of the encoding sequence in the genome.

3 Use cases

Annotating RNA-seq, raw reads



- (Short) RNA reads
- 👉 map to genome
- 👉 count number of reads within exons of genes

Annotation RNA-seq, raw reads

- *example* (Love et al. 2015):
  - bulk RNA-seq data
  - alignment against genome using STAR

```
library(airway)
dir(system.file("extdata", package = "airway"))
```

```
 [1] "GSE52778_series_matrix.txt"        "Homo_sapiens.GRCh37.75_subset.gtf"
 [3] "quants"                            "sample_table.csv"
 [5] "SraRunInfo_SRP033351.csv"          "SRR1039508_subset.bam"
 [7] "SRR1039509_subset.bam"             "SRR1039512_subset.bam"
 [9] "SRR1039513_subset.bam"             "SRR1039516_subset.bam"
[11] "SRR1039517_subset.bam"             "SRR1039520_subset.bam"
[13] "SRR1039521_subset.bam"
```

- *task*: quantifying reads per gene.
- *required*: positional (genome) information for genes (exons)
- 👉 get annotation resource for genome version / Ensembl release.

```
library(EnsDb.Hsapiens.v75)
edb <- EnsDb.Hsapiens.v75
```

- Get positional information for exons

```r
exns <- exonsBy(edb, by = "gene")
exns
```

```
GRangesList object of length 64102:
$ENSG00000000003
GRanges object with 17 ranges and 1 metadata column:
       seqnames            ranges strand |         exon_id
          <Rle>         <IRanges>  <Rle> |     <character>
   [1]        X 99894942-99894988      - | ENSE00001828996
   [2]        X 99891790-99892101      - | ENSE00001863395
   ...      ...               ...    ... .             ...
  [16]        X 99885756-99885863      - | ENSE00000868868
  [17]        X 99883667-99884983      - | ENSE00001459322
  -------
  seqinfo: 273 sequences (1 circular) from GRCh37 genome

...
<64101 more elements>
```

- Get the BAM files with the alignment results for the present experiment.

```r
#' Get the BAM file names in the *airway* package
f <- dir(system.file("extdata", package = "airway"),
         pattern = "bam$", full.names = TRUE)

library("Rsamtools")
bamfiles <- BamFileList(f, yieldSize = 2000000)
```

- Use `GenomicAlignments::summarizeOverlaps()` to count reads falling within exon boundaries.

```r
library(GenomicAlignments)
```

```r
se <- summarizeOverlaps(features = exns, reads=bamfiles,
                        mode = "Union", singleEnd = FALSE,
                        ignore.strand = TRUE,
                        fragments = TRUE)
se
```

```
class: RangedSummarizedExperiment
dim: 64102 8
metadata(0):
assays(1): counts
rownames(64102): ENSG00000000003 ENSG00000000005 ... LRG_98 LRG_99
rowData names(0):
colnames(8): SRR1039508_subset.bam SRR1039509_subset.bam ...
  SRR1039520_subset.bam SRR1039521_subset.bam
colData names(0):
```

- Result: gene count data.
- ℹ️ afternoon labs will use an updated workflow using *Salmon* for alignment against the transcriptome and the *tximeta* Bioconductor package for managing data and annotation resources.

Annotating gene information

- Common task in annotation: add additional metadata to existing identifiers.

- *example*: bulk RNA-seq data *airway*:

```
se
```

```
class: RangedSummarizedExperiment
dim: 64102 8
metadata(0):
assays(1): counts
rownames(64102): ENSG00000000003 ENSG00000000005 ... LRG_98 LRG_99
rowData names(0):
colnames(8): SRR1039508_subset.bam SRR1039509_subset.bam ...
  SRR1039520_subset.bam SRR1039521_subset.bam
colData names(0):
```

ℹ️ gene quantification data, 8 samples.

- What gene identifiers do we have?

```
#' extract gene identifiers
ids <- rownames(se)
head(ids)
```

```
[1] "ENSG00000000003" "ENSG00000000005" "ENSG00000000419" "ENSG00000000457"
[5] "ENSG00000000460" "ENSG00000000938"
```

- We've got Ensembl gene IDs.
- 🎯 get official HGNC gene symbols (names) the genes.

- 👉 use `AnnotationDbi::mapIds()` with *org.Hs.eg.db*.

```r
library(org.Hs.eg.db)
#' available gene annotations:
columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

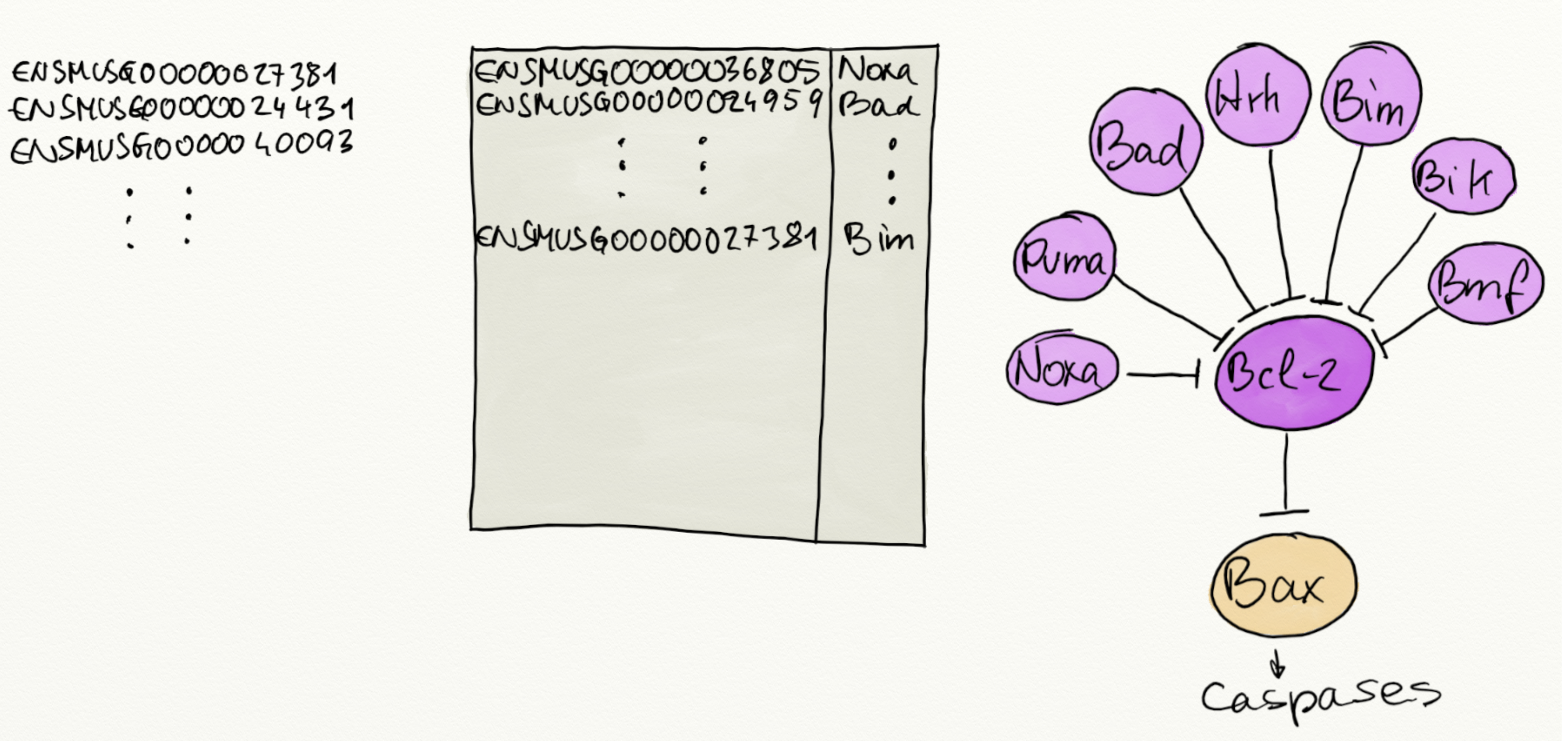
```r
#' map Ensembl gene IDs to official gene symbols
symbols <- mapIds(org.Hs.eg.db, keys = ids, keytype = "ENSEMBL",
                  column = "SYMBOL", multiVals = "first")
head(symbols)
```

```
ENSG00000000003 ENSG00000000005 ENSG00000000419 ENSG00000000457 ENSG00000000460
       "TSPAN6"          "TNMD"          "DPM1"         "SCYL3"         "FIRRM"
ENSG00000000938
          "FGR"
```

ℹ️ `multiVals`… to handle ambiguity; default is `"first"` but there are other options too.

Annotation for enrichment analysis

- going one step further ... 🎯 annotating genes to biological pathways.



- 👉 define gene sets.
- 👉 map identifiers to gene identifiers used in pathway annotation resource.

Annotation for enrichment analysis

- Enrichment analysis for biological pathways from reactome.
- *example*: use the *reactome.db* package to get the list of genes per pathway.

```
library(reactome.db)

#' Available data
columns(reactome.db)
```

```
[1] "ENTREZID"   "GO"         "PATHID"     "PATHNAME"   "REACTOMEID"
```

ℹ️ gene identifiers: resource uses NCBI EntrezGene IDs.

- Define the gene sets.

```
mapping <- select(reactome.db, keys = keys(reactome.db),
                  columns = c("ENTREZID", "PATHID"))
head(mapping)
```

```
  ENTREZID        PATHID
1        1  R-HSA-109582
2        1  R-HSA-114608
3        1  R-HSA-168249
4        1  R-HSA-168256
5        1 R-HSA-6798695
6        1   R-HSA-76002
```

- Convert into a `list` of genes per pathway.

```r
gs <- split(mapping$ENTREZID, mapping$PATHID)
gs[1:2]
```

```
$`R-BTA-1059683`
[1] "280826" "282081" "507359" "512484" "527418" "533590"

$`R-BTA-109581`
 [1] "100101492" "100140945" "100296226" "280730"    "280955"    "281020"
 [7] "281048"    "281169"    "282125"    "282126"    "282152"    "282321"
[13] "282691"    "286862"    "286863"    "287022"    "327672"    "404144"
[19] "404151"    "407101"    "407111"    "408016"    "493720"    "493999"
[25] "504707"    "504727"    "506223"    "507481"    "507804"    "507981"
[31] "508345"    "508646"    "510373"    "510767"    "512405"    "514090"
[37] "516952"    "517850"    "528453"    "529902"    "531514"    "533233"
[43] "533527"    "533949"    "538801"    "539003"    "539350"    "539679"
[49] "539941"    "540108"    "540369"    "540444"    "540643"    "540892"
[55] "541141"    "614840"    "616398"    "768311"    "785911"
```

- Map input genes from Ensembl to EntrezGene IDs.

```r
entrez <- mapIds(edb, keys = rownames(se), keytype = "GENEID",
                 column = "ENTREZID", multiVals = "first")
```

- 👉 use this definition of gene sets in enrichment functions, such as `EnrichmentBrowser::sbea()` or others.

```r
library(EnrichmentBrowser)

rownames(se) <- entrez
#' ... assuming differential expression analysis was done too ...
res <- sbea(method = "ora", se, gs = gs)
```

Annotating mass spectrometry data

- … 👀 Thursday’s Metabolomics lecture …

Annotating single-cell RNA-seq

🎯 cell type recognition from single-cell RNA-seq data.

SingleR package:

👉 reference gene expression data for known (pure) cell types.

👉 unbiased annotation by comparing single-cell gene expression against reference.

- Load reference dataset using *celldex*: Human Primary Cell Atlas

```r
library(celldex)
hpca_se <- HumanPrimaryCellAtlasData()
hpca_se
```

```
class: SummarizedExperiment
dim: 19363 713
metadata(0):
assays(1): logcounts
rownames(19363): A1BG A1BG-AS1 ... ZZEF1 ZZZ3
rowData names(0):
colnames(713): GSM112490 GSM112491 ... GSM92233 GSM92234
colData names(3): label.main label.fine label.ont
```

- Test data set: (La Manno et al. 2016), subset to first 100 cells

```r
library(scRNAseq)
hESCs ← LaMannoBrainData("human-es")
hESCs ← hESCs[, 1:100]
```

- Annotate each cell based on (non-parametric) correlation of gene expression data between test and reference.

```r
library(SingleR)
pred_ct ← SingleR(test = hESCs, ref = hpca_se, assay.type.test = 1,
                  labels = hpca_se$label.main)
pred_ct
```

```
DataFrame with 100 rows and 4 columns
                                              scores                labels delta.next
                                            <matrix>           <character>  <numeric>
1772122_301_C02 0.347652:0.139036:0.109547: ... Neuroepithelial_cell  0.0833286
1772122_180_E05 0.361187:0.155395:0.134934: ...              Neurons  0.0728350
...                                              ...                  ...        ...
1772122_298_F09 0.332361:0.173357:0.141439: ... Neuroepithelial_cell  0.1200606
1772122_302_A11 0.324928:0.127518:0.101609: ...            Astrocyte  0.0509478
                      pruned.labels
                        <character>
1772122_301_C02 Neuroepithelial_cell
1772122_180_E05              Neurons
...                               ...
1772122_298_F09 Neuroepithelial_cell
1772122_302_A11            Astrocyte
```

Thank you for your attention 🙌

References

La Manno, Gioele, Daniel Gyllborg, Simone Codeluppi, et al. 2016. "Molecular Diversity of Midbrain Development in Mouse, Human, and Stem Cells." *Cell* 167 (2): 566–580.e19. https://doi.org/10.1016/j.cell.2016.09.027.

Love, Michael I., Simon Anders, Vladislav Kim, and Wolfgang Huber. 2015. "RNA-Seq Workflow: Gene-Level Exploratory Analysis and Differential Expression." … 4: 1070. https://doi.org/10.12688/f1000research.7035.1.